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A critical review of prospects and operational challenges of microaeration and iron dosing for in-situ biogas desulfurization

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ABSTRACT

Anaerobic digestion is a widely used waste valorization method that produces methane-rich biogas. Notably, biogas upgrading to renewable natural gas has attracted considerable attention. However, extensive desulfurization is required for its upgrading due to the presence of various impurities like hydrogen sulfide and volatile organosulfur compounds in biogas. Recently, in-situ biogas desulfurization technologies, particularly microaeration and iron dosing, have received significant attention, with a few successful full-scale demonstrations. A few reports suggested that these two technologies can be economically competitive with ex-situ biogas desulfurization technologies. This article comprehensively reviews the in-situ desulfurization efficiency of these processes during AD. This review covers mechanisms and desulfurization pathways, process parameters, process control, and operational challenges of these technologies. Moreover, the economic feasibility of microaeration and iron dosing approaches were highlighted. Finally, based on the literature review, key research gaps, challenges, and future research perspectives are outlined to guide the prospects for future studies.

1. Introduction

Anaerobic digestion (AD) is one of the most popular biological processes for waste valorization (Azizi et al., 2022; Barua and Dhar, 2017; Ryue et al., 2020; Zhou et al., 2021). The AD produces renewable methane-rich biogas and reduces pathogen content and the final volume of the sludge prior to disposal. Biogas can be utilized to co-generate heat and electricity or upgraded to renewable natural gas (RNG). Notably, biogas upgrading to RNG received significant momentum in recent years (Ncube et al., 2021; Walker et al., 2018). However, biogas upgrading to RNG requires extensive purification to remove carbon dioxide (CO_2), hydrogen sulfide (H_2S), and other impurities (Das et al., 2022; Dykstra and Pavlostathis, 2021; Zakaria et al., 2020). Other impurities include various volatile organosulfur compounds (methyl mercaptan, dimethyl sulfide, dimethyl disulfide, etc.) and volatile organic compounds (e.g., siloxanes) (Das et al., 2022; Dhar et al., 2011a, 2011b, 2012; Žák et al., 2018). H_2S and volatile organosulfur compounds in biogas are detrimental to metal pipes and biogas utilization equipment. During biogas combustion, H_2S conversion to sulfur oxides and sulfuric acid results in severe corrosion of process equipment (Andriani et al., 2020). H_2S levels in biogas should be <1000 ppm for utilization in traditional boilers and < 0.1 ppm for internal combustion engines (Giordano et al., 2019).

During the AD process, sulfur-containing compounds in the feedstocks are reduced to H_2S and other volatile sulfur compounds (VSCs) (Novak et al., 2002). For instance, sulfate-reducing bacteria (SRB) can reduce the dissolved sulfate to dissolved sulfide (HS^-). Sulfate, sulfite, and thiosulfate can serve as electron acceptors for SRB, consuming organic acids and producing sulfide. HS^- may be converted to dissolve H_2S. Concentrations of HS^- and H_2S in the aqueous phase are defined according to the Henderson–Hasselbach equation (Okoro and Sun, 2019); the concentrations of HS^- and H_2S in digester change with pH. Also, studies have shown that the degradation of organic macromolecules, such as proteins, is closely related to the production of VSCs (Novak et al., 2002). Amino acids are the basic structural building block of proteins. The production of VSCs starts with the degradation of proteins to produce sulfur-containing amino acids such as cysteine, methionine, etc. Under anaerobic conditions, these sulfur-containing amino acids can produce H_2S and mercaptans, such as methyl mercaptan or methanethiol (Yoshimura et al., 2000).

The extensive desulfurization requirements may substantially

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increase the biogas upgrading costs for its various utilization options. Various available ex-situ and in-situ technologies are available for biogas desulfurization. In the ex-situ processes, the biogas is cleaned in a separate downstream unit using physicochemical or biological processes, such as scrubbing, biofilter, biotrickling filters, absorption, and adsorption (Okoro and Sun, 2019). On the other hand, the in-situ processes are localized within the digester and limit dissolved sulfide conversion to H_2S (Okoro and Sun, 2019). Many commercial ex-situ technologies are available in the market, while in-situ processes are often considered relatively underdeveloped compared to ex-situ processes due to the lack of adequate information available on full-scale demonstration experience (Sarker et al., 2018). Nonetheless, it has been reported that in-situ desulfurization processes (microaeration and iron dosing) can result in significantly higher cost-saving than ex-situ technologies (particularly physicochemical processes) (Okoro and Sun, 2019; Sarker et al., 2018). Moreover, microaeration and iron dosing are getting more interest due to a few successful full-scale demonstrations in recent years (Erdirencelebi and Kucukhemek, 2018; Giordano et al., 2019; Jeníček et al., 2017; Kobayashi et al., 2012; Kulandaivelu et al., 2020).

However, there is a lack of a comprehensive review of lab, pilot, and full-scale studies on these two technologies, encompassing different fundamental and technical aspects and identifying further research needs. Consequently, the purpose of this article is to provide a critical review of available information related to two major in-situ biogas desulfurization technologies, microaeration and iron dosing. In addition to a synopsis of the fundamental mechanisms, a critical review of the desulfurization efficiencies, technology readiness level, process control, economic considerations, and operational challenges associated with each technology is provided. Based on the review, key knowledge gaps associated with these technologies are summarized, and future research perspectives relating to further developments are proposed. It should be noted that previous studies also outlined other potential benefits in terms of improving hydrolysis kinetics, process stability, and biogas production (Cai et al., 2018; Chen et al., 2020; Krayzelova et al., 2015; Nguyen and Khanal, 2018; Qiang et al., 2013). These aspects were not considered within the scope of this review. However, interested readers are referred to other review articles on these topics (Baek et al., 2019; Chen et al., 2020; Nguyen and Khanal, 2018).

2. Microaeration for biogas desulfurization

By tradition, exposure of anaerobic digesters to the aerobic condition has been strictly avoided to ensure the activity of the digester microbiome, which is usually composed of obligate anaerobic microbes. However, over the last two decades, various benefits of controlled injection of a little amount of air or pure oxygen into digesters have been reported (Chen et al., 2020; Krayzelova et al., 2015; Nguyen and Khanal, 2018). For instance, minimal aeration can provide high-quality biogas via scavenging sulfide, enhancing hydrolysis, and promoting a robust microbiome (anaerobes and facultative microbes) to ensure process stability and increased methane productivity from AD (Chen et al., 2020; Krayzelova et al., 2015; Nguyen and Khanal, 2018). Microaeration can be applied through different schemes by supplying air or pure oxygen in the liquid phase or the digester headspace (Krayzelova et al., 2015). Also, different aeration (air or pure oxygen) frequencies (intermittent and continuous) can be maintained (Krayzelova et al., 2015).

At low oxygen levels, sulfide can be biochemically oxidized to elemental sulfur and/or sulfate in most cases. Moreover, microaeration can enrich facultative hydrolytic and acidogenic microorganisms, thereby accelerating process kinetics and stability. As this review article focuses on the desulfurization of biogas, other features (i.e., improved hydrolysis and process stability) will not be discussed here. Despite 'microaeration' has been the most used term in the literature, other terminologies including limited aeration, microoxygenation, and moderate oxygenation have also been used to represent this method. For instance, researchers often used 'microoxygenation' when pure oxygen was used to create a micro-aerobic condition (Krayzelova et al., 2015).

2.1. Mechanisms and desulfurization pathways

The microaeration can facilitate the H_2S oxidation in both the gaseous and aqueous phases (Fig. 1), while the process integrates both chemical and biological reactions for H_2S removal (Chen et al., 2020; Kobayashi et al., 2012; Okoro and Sun, 2019). Sulfide serves as the electron donor, while oxygen is the terminal electron acceptor. Elemental sulfur (S) is widely reported as the most common product of biochemical oxidation; however, the oxidation product may vary depending on the H_2S to oxygen molar ratios (Chen et al., 2020; Kobayashi et al., 2012; Okoro and Sun, 2019). The reactions highlighting the different products are as follows (Chen et al., 2020; Kobayashi et al., 2012; Okoro and Sun, 2019):

$$H_2S + ½\ O_2 \rightarrow S + H_2O$$

$$H_2S + 2\ O_2 \rightarrow SO_4^{2-} + 2H^+$$

$$S + H_2O + 3/2\ O_2 \rightarrow SO_4^{2-} + 2H^+$$

$$2HS^- + O_2 \rightarrow 2S + 2OH^-$$

$$2HS^- + 4O_2 \rightarrow 2SO_4^{2-} + 2H^+$$

Moreover, the production of thiosulfate ($S_2O_3^{2-}$) from HS^- has also been reported (Chen et al., 2020; Kobayashi et al., 2012; Okoro and Sun, 2019).

$$2HS^- + O_2 \rightarrow S_2O_3^{2-} + H_2O$$

The formation of elemental sulfur requires a relatively lower amount of oxygen (0.5 mol O_2/mol S^{2-}) than sulfate formation (2 mol O_2/mol S^{2-}) (Krayzelova et al., 2015), making it more attractive from an economic perspective. However, the production of elemental sulfur or sulfate usually depends on the oxygen dosing; thus, elemental sulfur should be the major product under oxygen-limited (i.e., microaerobic) conditions. In general, sulfate can be formed under sulfide-limiting conditions (Krayzelova et al., 2015). For instance, Klok et al. (2013) reported increasing sulfide-oxidizing bacterial activity at sulfide concentrations of 0–0.15 mmol/L. At sulfide concentrations between 0.3 and 1.0 mmol/

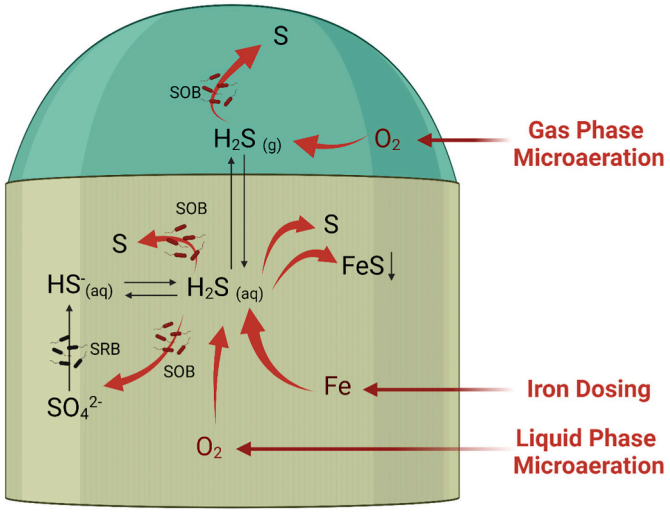


Fig. 1. Major in-situ sulfide removal pathways by microaeration and iron dosing.

L, biological activity reduced. Chemolithotrophic sulfide-oxidizing bacteria (SOB), such as *Acidithiobacillus thiooxidans*, *Thiomonas intermedia*, and *Thiomonas perometabolis*, were often reported as the key microorganisms involved in microaerobic desulfurization (Krayzelova et al., 2015). According to previous literature, chemical oxidation of sulfide to $S_2O_3^{2-}$ usually becomes critical when the activity of sulfide-oxidizing bacteria is limited, particularly when bioreactors are highly loaded with sulfide (Buisman et al., 1990; Krayzelova et al., 2015).

2.2. Desulfurization efficiency

As summarized in Table 1, several lab, pilot, and full-scale studies reported efficient (>90 % in most cases) H_2S removal with microaeration. Notably, Jeníček et al. (2017) recently published long-term (years 2003–2015) microaeration operation experiences from 7 full-scale anaerobic digesters (1600–30,000 m^3) in central Europe, which demonstrated >90 % desulfurization efficiencies for most facilities. In these facilities, microaeration systems were retrofitted within the sludge recirculation line without any modification in digester construction. Their microaeration systems consisted of an air compressor and the airflow control and monitoring unit, designed by K&K Klatovy, Czech Republic. Applying gas-phase microaeration in the digester headspace also showed high desulfurization efficiencies (68.2–99 %) in full-scale AD facilities treating sewage sludge and manure (Giordano et al., 2019; Kobayashi et al., 2012).

Giordano et al. (2019) reported 81 weeks of data on desulfurization efficiencies with microaeration (with air) in three full-scale thermophilic digesters. Despite high desulfurization efficiency (99 % H_2S removal from biogas), when air is used, removing residual oxygen and nitrogen would be vital, which could be quite expensive. Notably, the authors found that utilizing pure oxygen instead of air might be necessary to meet the European standard EN 16726 (CO_2, O_2, and H_2S content of 2.5–4 %, 0.001–1 %, and $\leq$ 5 ppm, respectively) of biomethane for pipeline utilization. Nonetheless, the electrical energy input for the oxygen-based microaeration operation estimated in their study (3.4 kWh/ton of biogas) was still lower than the microaeration cost estimated in another study that used oxygen for microaeration (Díaz et al., 2015). Thus, site-specific techno-economic assessment (e.g., air vs. oxygen microaeration, the post-processing requirement for residual O_2 and N_2 removal) would be necessary.

2.3. Process schemes

One of the key considerations in implementing microaeration is the injection point of the oxygen/air. Air/oxygen can be injected either into the gaseous phase (digester headspace or biogas recirculation line) or liquid phase (middle/top of the bioreactor or sludge recirculation line). Fig. 2 illustrates different dosing points of air/oxygen. However, among various schemes, dosing air/oxygen in the headspace in the digester is the most frequently used in previous studies (see Table 1). However, it should be noted that air injection in the sludge recirculation line was demonstrated in 7 full-scale anaerobic digesters in Europe (Jeníček et al., 2017). Over the long term (2003–2005), the demonstration has been successful, showing above 90 % H_2S removal for most digesters.

When air/oxygen is injected into the digester headspace, gaseous H_2S can directly react with oxygen. Thus, the oxygen requirement per given amount of H_2S will be minimized (Krayzelova et al., 2015). This is particularly important as lower air injection would also minimize biogas dilution by oxygen and nitrogen (Krayzelova et al., 2015). The residual oxygen and nitrogen content in the biogas must be controlled to meet the requirements of the downstream biogas utilization, as well as avoid methanogenic inhibition by a high level of residual oxygen (Krayzelova et al., 2015). When air is dosed into the liquid phase (i.e., sludge), co-oxidation of organics may be promoted, leading to oxygen losses and lower methane recovery. A report suggested that 13–59 % of O_2 can be consumed for non-specific oxidation of organics (Giordano et al., 2019). Thus, air/oxygen dosing requirements will be increased, which may also increase operating costs. However, another study reported that non-specific oxidation of organics would be negligible (COD balance was similar for control and microaerobic reactors) (Krayzelova et al., 2014).

Interestingly, Krayzelova et al. (2014) reported that when air/oxygen was injected in the liquid phase, dissolved sulfide removal was only

Table 1
Summary of microaeration studies focusing on desulfurization of anaerobic digester biogas.

Scale	Total Volume	Feedstock	Microaeration condition	Air dosing point	H_2S removal (Initial H_2S level)	Reference
Full-scale	1600–30,000 m^3	Sewage sludge	Air; 1–3 % of biogas	Sludge recirculation line	74–99 % (771–7580 ppm)	(Jeníček et al., 2017)
Full-scale	4500 m^3	Sewage sludge	Air; 0.2–2 % of biogas	Headspace	99 % (1200–2500 ppm)	(Giordano et al., 2019)
Full-scale	338 m^3	Manure	Air; 1 % of biogas production (20 m^3/d)	Headspace	68.2 % (2000–4000 ppm)	(Kobayashi et al., 2012)
Field-Pilot	8.5 m^3	Manure	Air; 1 % of digester volume	Headspace	99 % (2300–4000 ppm)	(Huertas et al., 2020)
Pilot	7 m^3	Sewage sludge	Pure oxygen (92–98 %); 5–34 L/m^3/d	Headspace, liquid, and biogas recirculation	99 % (500 ppm)	(Ramos et al., 2014c)
Pilot	250 L	Sewage sludge	Pure oxygen (4.4–6.2 L/m^3/d)	Sludge recirculation line	90 %	(Ramos and Fdz-Polanco, 2013)
Pilot	250 L	Sewage sludge	Pure oxygen; 0.2–0.25 m^3 O_2/m^3 feed	Headspace	97 % (15,811 ppm)	(Díaz et al., 2010)
Pilot	250 L	Sewage sludge	Air; 0.94–1.27 L O_2/L feed	Headspace	99 % (15,811 ppm)	(Díaz et al., 2010)
Pilot	250 L	Sewage sludge	Pure oxygen (frequently adjusted; O_2 of 0.02 %–0.20 % (v/v))	Headspace	100 %	(Ramos et al., 2014b)
Pilot	70 L	Sewage sludge	Pure oxygen (ORP level between −310 and −290 mV)	Liquid	99.5 % (6000 ppm)	(Nghiem et al., 2014)
Lab	15 L	Chicken manure	Air; 120–360 L/d	Headspace	99.7 % (5000–6000 ppm)	(Song et al., 2020)
Lab	20 L	Sludge	Pure oxygen; 4 mL/h	Headspace	97 % (5800 ppm)	(Andreides et al., 2021a)
Lab	4.5 L	Animal Wastewater	Air; 66.7 mL/L/min	Liquid	70.2 % (336 ppm)	(Yang and Deng, 2020)
Lab	20 L	Cheese whey solution	Air; 20 mL/h	Headspace	90 % (10,000 ppm)	(Andreides et al., 2021b)
Lab	10 L	Sewage sludge	Air; 1.6 L/day; O_2 1.7–9.2 % of biogas	Liquid	99 %	(Jenicek et al., 2014)
Lab	11 L	Sewage sludge	Air; 1.1 L/day	Sludge recirculation line	92 %	(Jenicek et al., 2008)

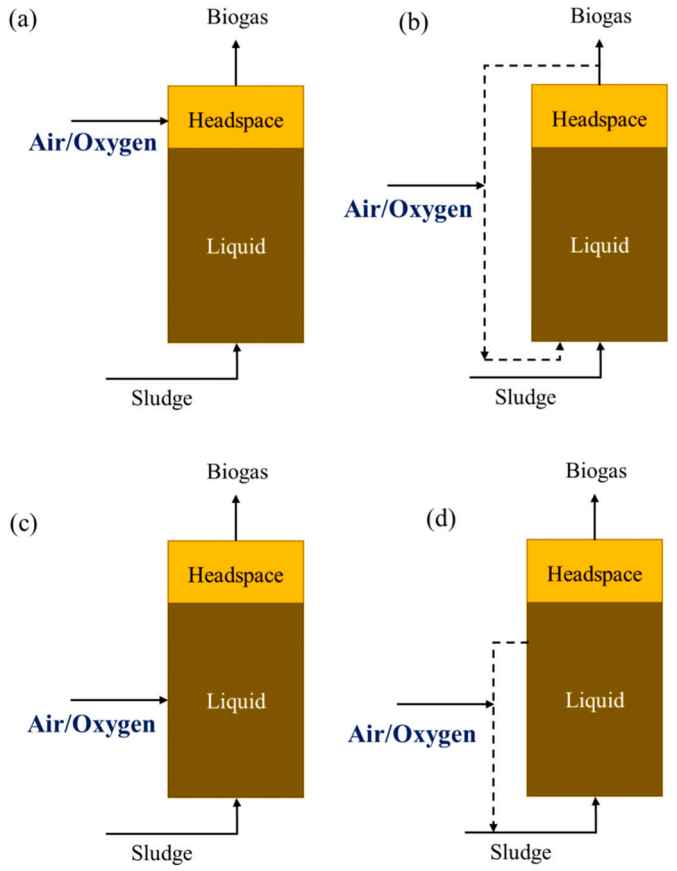


Fig. 2. Conceptual schematic of possible process schemes for incorporating microaeration in anaerobic digesters via injecting air/oxygen in (a) headspace, (b) biogas recirculation line, (c) liquid phase, (d) sludge recirculation line.

20–30 %, while H_2S removal in biogas was as high as 79 %. Thus, it was evident that H_2S oxidation also occurs in the headspace, even when air is injected into the liquid phase. Ramos et al. (2014a) studied the effect of different headspace volumes (25, 0.3, and 0 L) on biogas desulfurization with a pilot-scale (266 L of total volume) anaerobic digester. In their study, oxygen was injected into the liquid phase. The authors found that complete biogas desulfurization could be achieved with 25 L of headspace. In contrast, the headspace of <0.3 L resulted in minimal or no desulfurization despite similar oxygen being supplied to the digester. Thus, their results suggested that the available surface area in the digester headspace would be critical to efficiently desulfurize biogas (Ramos et al., 2014a). However, keeping more headspaces would undoubtedly reduce the digester working volume, which can ultimately impact the overall process economics and needs further attention.

Several studies reported that sulfide-oxidizing bacterial population could grow on the walls of the digester headspace (Díaz et al., 2011; Kobayashi et al., 2012; Ramos et al., 2014a, 2014b; Rodríguez et al., 2012). Furthermore, these studies also reported the deposition of elemental sulfur in the headspace walls. In contrast, a few studies suggested minimal elemental sulfur deposition in headspace when air/oxygen is supplied in fine bubbles (Khanal and Huang, 2003, 2006). Nonetheless, the deposition of elemental sulfur seems to be the most common phenomenon. It has been suggested that the deposition of elemental sulfur on the digester headspace walls and biogas pipes can result in clogging issues, resulting in headspace overpressure. Therefore, Ramos et al. (2014b) suggested a cleaning interval of <14 months would be necessary for digester operation with microaeration. However, further research is needed to develop a comprehensive guideline regarding the cleaning method and frequency, estimate associated maintenance costs, or develop a potential operational strategy to avoid sulfur deposition on the headspace walls.

To date no studies systematically compared four schemes (Fig. 2) for applying microaeration for in-situ desulfurization. However, as evident from the results summarized in Table 1, most studies reported high desulfurization efficiency (≥90 %) irrespective of the process schemes used. Thus, holistic consideration of various factors, such as co-oxidation of organics, capital/operational costs, digester modifications, and maintenance requirements, is required to select an optimum process scheme.

2.4. Process control

In the literature, the amount of oxygen supplied for the microaerobic condition was expressed in different ways (see Table 1), including air/oxygen supplied per unit volume of the reactor per day ($L/L_{reactor}$-day), air/oxygen supplied per unit volume of feedstock (L/L feed), oxygen to biogas ratio (% of biogas), and volumetric flow rate of air/oxygen (L/day). Thus, there has been no standardized approach to represent the micro-aerobic condition in the reported studies.

Unlike the conventional aerobic process, the dissolved oxygen (DO) concentration cannot be used as a control parameter for the microaeration process. Under the microaerobic condition, DO concentration ranges <1 mg/L (Chen et al., 2020; Krayzelova et al., 2015). Particularly, elemental sulfur or sulfate formation from sulfide proceeds at DO concentrations <0.1 mg/L (Chen et al., 2020; Krayzelova et al., 2015), which is too low to be reliably measured with a traditional DO meter. Therefore, redox or oxidation-reduction potential (ORP) has been used as an alternative control parameter for microaeration (Chen et al., 2020; Nguyen and Khanal, 2018).

ORP can be defined as the tendency of an aqueous solution to accept or donate electrons and is frequently used to measure the oxidative (more positive values) or reductive (more negative values) environment of the liquid phase (Chen et al., 2020; Nguyen and Khanal, 2018). ORP changes linearly with the logarithmic DO concentration. Thus, the ORP electrode can more accurately sense tiny changes in DO concentrations (Chen et al., 2020; Nguyen and Khanal, 2018). For example, an ORP of −50 mV corresponds to the 0.1 mg/L DO probe, while ORP values become more negative with further reduction in DO level (Chen et al., 2020; Nguyen and Khanal, 2018). Therefore, the ORP-based control strategy has been reported in the literature. For instance, Khanal and Huang (2003) proposed an ORP-based oxygen dosing strategy for microaerobic desulfurization during AD, where oxygen dosing was automatically turned on when the ORP dropped 10 mV below the target value. Then, oxygen dosing was provided until ORP increased 10 mV above the target value. ORP values in microaerobic environment range from 0 to −300 mV (vs. standard hydrogen electrode (SHE)) (Chen et al., 2020; Nguyen and Khanal, 2018). The average ORP level in anaerobic digesters is −300 mV or lower. However, a wide range of ORP values has been reported in the literature for microaerobic AD, ranging from −460 mV to higher than −150 mV (Chen et al., 2020; Nguyen and Khanal, 2018). Such variations could be attributed to different operating conditions used in different studies. Furthermore, whether the ORP values are reported based on a SHE as a reference or another reference electrode is often unclear. Therefore, a site-specific ORP-based oxygen dosing strategy is needed.

Air/oxygen dosing rate can also be manipulated to adjust the changes in biogas H_2S levels and biogas flow rates using a proportional–integral–derivative (PID) controller (Krayzelova et al., 2015; Ramos and Fdz-Polanco, 2013). Air/oxygen flow rate can be set based on the difference between the measurement and the target H_2S level. Some studies also reported air/oxygen dosing based on the flow rate of biogas. For instance, providing an air supply of 1–3 % of biogas was required to remove ~90 % H_2S (771–7580 ppm) from biogas in full-scale digesters, where the air was injected into the sludge recirculation line (Jeníček et al., 2017). For injection in the headspace, an air supply of 0.2–2 % of biogas could provide 99 % H_2S removal (1200–2500 to <10 ppm) from

biogas in full-scale sludge anaerobic digester (Giordano et al., 2019).

2.5. Operational challenges

The interest in microaeration as a biogas desulfurization method removal has been steadily growing over the last two decades. Notably, its successful demonstrations in full-scale AD facilities are encouraging (Giordano et al., 2019; Jeníček et al., 2017; Kobayashi et al., 2012). Nonetheless, previous studies do not provide clear guidelines on process control parameters (e.g., optimum ORP, oxygen/air flow rate). Without appropriate process optimization, excess microaeration may lead to dilution of biogas and process instability due to the deterioration of the methanogenic activity.

Furthermore, other operational challenges, such as maintenance requirements, require further attention. The deposition of elemental sulfur on the headspace can eventually create clogging issues in biogas pipes and digester walls. Therefore, regular cleaning would ensure high oxygen transfer efficiency and reduce microaeration operational costs (Ramos et al., 2014b). Several studies highlighted that biofilms of the sulfide-oxidizing bacterial population on the digester headspace walls would be critical to effective desulfurization (Díaz et al., 2011; Kobayashi et al., 2012; Ramos et al., 2014a; Rodríguez et al., 2012). It is expected that sulfide-oxidizing bacterial biofilms will be lost during the cleaning of the headspace walls for elemental sulfur. Thus, reestablishing the sulfide-oxidizing bacterial biofilms would be necessary to attain the desired desulfurization efficiency. However, based on an extensive literature search, no reports could be found on the time required to enrich the sulfide-oxidizing microbiome after cleaning. The liquid-phase microaeration can possibly cause less clogging because less oxygen reaches the gaseous phases; however, it will result in non-specific oxidation of organics, leading to low methane yield (Jeníček et al., 2017; Krayzelova et al., 2015).

Although a small amount of oxygen/air is dosed during microaeration, explosion risks of methane/oxygen mixture should not be overlooked (Krayzelova et al., 2015). Moreover, the dilution of biogas by residual O_2 and N_2 has been identified as another challenge in microaeration operation (Giordano et al., 2019; Krayzelova et al., 2015). It has been suggested that residual oxygen concentration in biogas can be used as a controlling parameter for microaeration with pure oxygen (Giordano et al., 2019); however, N_2 would still be unavoidable when air is used for microaeration. The dilution of biogas may pose the need for biogas post-processing for oxygen and nitrogen removal. Therefore, future research should develop an optimum operational strategy to avoid biogas dilution by microaeration.

3. Iron dosing for biogas desulfurization

Iron salts/oxides dosing in digesters has been reported as another simple and often cost-effective approach for in-situ mitigation of H_2S generation during AD (Okoro and Sun, 2019). Iron salts (ferrous or ferric) and iron oxides can react with dissolved sulfide to form insoluble iron sulfide precipitates (Fig. 1). Consequently, the stripping of H_2S into the biogas can be prevented. However, compared to direct dosing of iron in digesters or into the digester influent, iron has been widely implemented for sulfide removal in sewer systems (Firer et al., 2008; Gutierrez et al., 2010; Rebosura et al., 2020; Zhang et al., 2009), which also demonstrated a positive effect in mitigating H_2S generation during AD (Ge et al., 2013; Kulandaivelu et al., 2020; Rebosura et al., 2018). For instance, Ge et al. (2013) reported that iron addition to the sewer systems at a relatively low concentration (5 mg/L) could be adequate to provide complete H_2S removal from AD biogas, which would provide an excellent corrosion management solution and save in maintenance cost of biogas-utilizing equipment. This approach has been widely implemented in Australia (Ganigue et al., 2011).

In addition to in-situ desulfurization, many studies investigated iron dosing in anaerobic digesters to enhance methanogenesis by improving enzymatic activity (Cai et al., 2018; Qiang et al., 2013). Notably, Fe also encompasses numerous enzymes and co-enzymes associated with the methanogenesis process (Cai et al., 2018; Qiang et al., 2013). Also, iron oxides, such as magnetite and hematite, have been used to promote electrical syntrophy between bacteria and archaea, improving process kinetics (Barua and Dhar, 2017). However, this review primarily focused on the studies related to biogas desulfurization.

3.1. Desulfurization mechanisms

In general, divalent (Fe^{2+}) and trivalent (Fe^{3+}) iron salts are the most commonly used for the precipitation of sulfides (Nägele et al., 2017; Okoro and Sun, 2019). For instance, iron (II) chloride ($FeCl_2$), iron (III) chloride ($FeCl_3$), iron (III) oxide-hydroxide (FeOOH), and iron (III) hydroxide ($Fe(OH)_3$) can facilitate sulfide precipitation via the following reactions (Nägele et al., 2017; Okoro and Sun, 2019).

$$FeCl_2 + HS^- + H^+ \rightarrow FeS + 2HCl$$

$$FeCl_3 + 2HS^- + H^+ \rightarrow FeS + 3HCl + S$$

$$2FeOOH + 3HS^- + 3H^+ \rightarrow 2FeS + S + 4H_2O$$

$$2Fe(OH)_3 + 3HS^- + 3H^+ \rightarrow 2FeS + S + 6H_2O$$

Although estimating the required iron dosages for sulfide precipitation using reaction stoichiometry seems quite straightforward, it can still be challenging to achieve desired desulphurization with stoichiometric dosages due to the co-precipitation of other compounds (Okoro and Sun, 2019). For instance, iron can bind with phosphate to form ferrous phosphate precipitates, such as vivianite ($Fe_3(PO_4)_2.8H_2O$) and other ferrous-(hydroxyl)-phosphate compounds (Ge et al., 2013; Roussel and Carliell-Marquet, 2016). The following equation (Eq. (1)) has been used by multiple studies to estimate the required iron dosages for target desulfurization efficiency (Nägele et al., 2017; Okoro and Sun, 2019).

$$Fe = \beta \times (M_{Fe}/M_S) \times \left[\left(\left(H_2S_{(aq)}/f_{H2S}\right) \times V_{Substrate}\right) + \left(\left(\Delta H_2S \times \rho H_2S \times V_{biogas}\right)/1000\right)\right] \quad (1)$$

where, Fe is the amount of iron supplied in g/d, $H_2S_{(aq)}$ is the dissolved sulfide in g/m^3, f_{H2S} is the mass fraction of total dissolved sulfur present as dissolved H_2S, ΔH_2S is the H_2S removal from biogas in ppm, $V_{Substrate}$ is influent flow rate in m^3/d, V_{biogas} is the biogas flowrate in m^3/d, ρH_2S is the density of H_2S in g/L under prevailing temperature and pressure, M_{Fe} is the molar mass of iron in g/mol, M_S is the molar mass of sulfur in g/mol, and β is the correction factor to account for over-dosing (range from 1.7 to 5), since the iron may react with other compounds (e.g., phosphates) that may be present in the digester.

3.2. Desulfurization efficiency

Table 2 summarizes previous studies focusing on in-situ biogas desulfurization in AD by iron dosing. Many studies reported complete desulfurization, while others reported moderate (~50 %) H_2S removal from biogas (see Table 2). Notably, full-scale demonstrations of iron-based biogas desulfurization have been reported (Erdirencelebi and Kucukhemek, 2018; Kulandaivelu et al., 2020). Kulandaivelu et al. (2020) recently reported a full-scale demonstration of iron ($FeCl_2$) dosing in wastewater treatment plant inlet works (Trial-1) and upstream sewer network (Trial-2) in the Luggage point-wastewater treatment plant in Brisbane, Queensland, Australia. During Trial-1 (dosing in inlet works), elevated iron concentration in the AD influent led to a 77.6 ± 3.6 % decrease in dissolved sulfide level; however, H_2S level in the biogas decreased by 36 % (303.7 to 194.4 ppm). During Trial-2 (dosing in the upstream sewer network, ~18 km away from the treatment plant), the H_2S level decreased by 45.2 %. However, based on previous lab experience with $FeCl_3$ (Rebosura et al., 2018), the authors suggested

Table 2
Summary of studies focusing on in-situ biogas desulfurization in anaerobic digestion by iron dosing.

Volume of digester	Feedstock	Iron	Dosage	Maximum H_2S removal	Reference
1 L	Sludge	$FeCl_3$	4000–4500 mg /kg TS	93 %	(Akgul et al., 2017)
- (Full-scale)	Sludge	$FeCl_2$	16.9–21.6 mg Fe/g dry sludge	36–45 %	(Kulandaivelu et al., 2020)
7000 m^3 (Full-scale)	Sludge	$FeCl_3$	24–105 mg/L	72 %	(Erdirencelebi and Kucukhemek, 2018)
1 L	Manure	Fe_3O_4	100–1000 mg/L	77.24 %	(Farghali et al., 2020)
1 L	Manure	Fe_2O_3	1000–1000 mg/L	53.52 %	(Farghali et al., 2020)
16 L	Manure + Energy crops	FeOOH (powder)	-[a]	73.10 %	(Nägele et al., 2017)
16 L	Manure + Energy crops	FeOOH (granular)	-[a]	69.42 %	(Nägele et al., 2017)
16 L	Manure + Energy crops	$FeCl_2$	-[a]	68.18 %	(Nägele et al., 2017)
16 L	Manure + Energy crops	$Fe(OH)_3$	-[a]	57.83 %	(Nägele et al., 2017)
5 L	Manure	$FeCl_2$	0–32 mmol/L	98.9 %	(Jiang et al., 2017)
5 L	Manure	$FeCl_3$	0–32 mmol/L	99 %	(Jiang et al., 2017)
5 L	Manure	$Fe(OH)_3$	0–32 mmol/L	99.1 %	(Jiang et al., 2017)
5 L	Manure	Fe_2O_3	0–32 mmol/L	90.5 %	(Jiang et al., 2017)
5 L	Manure	$FeSO_4$	0–32 mmol/L	56.7 %	(Jiang et al., 2017)
4.6 L	Sludge	$FeCl_3$	0.8–1.5 mol Fe/mol S	94 %	(Cao et al., 2019)
–	Sludge	$FeCl_3$	27 mg Fe/L	99 %	(Charles et al., 2006)

[a] Calculated using Eq. (1).

that possibly increasing iron dosing could provide higher H_2S removal in biogas (Kulandaivelu et al., 2020). Nonetheless, iron dosing had a trivial impact on methane production while improving sludge dewaterability and phosphorous removal (Kulandaivelu et al., 2020). Considering the cost of $FeCl_2$ at $340/ton, during Trial-1, the treatment cost was estimated at $0.06/$m^3$ of wastewater. In contrast, during Trial-2, the treatment cost was $0.01/$m^3$ of wastewater. However, the authors suggested that direct dosing of $FeCl_2$ to the digester (to maintain the same Fe level in the digester as Trial-2) can save chemical costs up to $360/day.

Recently, another report was published on the full-scale demonstration of $FeCl_3$ dosing (24–105 mg/L) in anaerobic digesters (total flow rate of 1400 m^3/d of sludge) in a wastewater treatment plant in Turkey (Erdirencelebi and Kucukhemek, 2018). The data from the 17-months demonstration period showed up to 72 % H_2S removal from biogas. The authors also suggested that the digesters stabilized within 11 days after the iron dosing waste started. Thus, there is a possibility that the effectiveness of iron addition can be tested without disturbing the operation of the AD facility.

3.3. Comparison of various iron compounds

To date, a wide range of iron compounds has been explored for in-situ biogas desulfurization (see Table 2). Although the majority of studies used divalent and trivalent iron salts, limited information is available on the systematic comparison of these compounds. Based on the extensive literature search, a published report could be found on the comparison of FeOOH, $FeCl_2$, and $Fe(OH)_3$ for H_2S removal during lab-scale anaerobic co-digestion of manure and energy crops (Nägele et al., 2017). Although $FeCl_2$ showed the fastest response, both granular and powdered forms of FeOOH showed slightly higher or quite comparable desulfurization efficiencies (69.42–73.10 % vs. 68.1 % H_2S removal). In contrast, $Fe(OH)_3$ demonstrated the lowest desulfurization efficiencies. Although the study did not reveal the underlying mechanisms, it appeared that the powder form of FeOOH provided slightly better performance than the granular FeOOH, while the surface area of powdered FeOOH was 10-times higher than the granular form. Interestingly, $FeCl_2$, added in its liquid form, demonstrated the fastest response, indicating its suitability for quick desulfurization. Erdirencelebi and Kucukhemek (2018) also reported that the time for response to the $FeCl_3$ (41–42 % aqueous solution) dosing was as low as 11 days for full-scale sludge anaerobic digesters.

Between $FeCl_2$ and $FeCl_3$, $FeCl_3$ was frequently endorsed by researchers (Charles et al., 2006; Ge et al., 2013; Kiilerich et al., 2018; Kulandaivelu et al., 2020). This is due to the fact that ferrous salts (Fe^{2+}) react directly with dissolved sulfide forming ferrous sulfide precipitates (FeS), while ferric ions (Fe^{3+}) oxidize dissolved sulfide to elemental sulfur, with itself being reduced to Fe^{2+}, which subsequently precipitates sulfide (Kulandaivelu et al., 2020). Kiilerich et al. (2018) demonstrated that dissolved sulfide precipitation rates would be faster with Fe(III) than with Fe(II). Moreover, as compared to Fe(II), Fe(III) demonstrated good sulfide precipitation efficiencies over a wide pH range. Based on the extensive literature search, limited information could be found on the systematic comparison of different iron compounds linking their characteristics, such as solubility, specific surface area, and surface morphology. However, these characteristics can also be critical in their effectiveness in in-situ desulfurization. For instance, in a study by Zhou et al. (2016), among different iron ores, Limonite was more effective for H_2S control in biogas than Hematite due to its relatively higher surface area (115.48 vs. 3.63 m^2/g) and pore volume (6.57×10^{-2} vs. 6.81×10^{-3} mL/g). Authors suggested that larger surface area and pore volume provided more contact surfaces for in-situ desulfurization reactions. Thus, these characteristics should be thoroughly investigated while comparing the effectiveness of different iron compounds for in-situ desulfurization.

3.4. Sustainable sources for iron

While iron dosing can be an easily retrofittable desulfurization process, the requirement of continuous dosing of iron still poses an economic barrier to its implementation. Interestingly, a few reports suggested the application of waste iron powder or scrap iron resulting from the cutting of iron and steel could also be utilized for biogas desulfurization (Andriamanohiarisoamanana et al., 2018; Farghali et al., 2020). To the best of our knowledge, Andriamanohiarisoamanana et al. (2018) first demonstrated that amendment of 2–8 g/L of waste iron powder (containing up to 85 % of Fe_3O_4) from a laser cutting machine could provide effective H_2S removal (93–99 %) during AD of dairy manure. Moreover, Farghali et al. (2020) studied the desulfurization performance of waste iron powder compared to commercial iron oxide nanoparticles (Fe_2O_3, 99 % purity, 20–40 nm particle size) during AD of manure. Different waste iron dosages (100–1000 mg/L) could provide H_2S removal in the range of 45.20 % to 77.24 %. In contrast, equivalent dosages of commercial iron oxide nanoparticles showed considerably lower desulfurization efficiencies (33.59 % to 53.52 % H_2S removal). Regarding underlying mechanisms, the authors postulated that the iron oxide core structure surrounding nano-iron might grow thicker as oxidation progresses, reducing interaction with sulfides.

Sun et al. (2015) suggested that iron-rich sludge collected from drinking water treatment plants can be utilized for sulfide control in the

sewer. Iron salts ($FeCl_3$) are often used as a coagulant in drinking water treatment plants to remove turbidity and natural organic material (NOM). Iron-rich sludge generated from drinking water treatment plants is mostly landfilled, while its utilization for sulfide removal can promote a circular economy approach. Salehin et al. (2020) demonstrated that in-sewer dosed iron could be recovered via a combination of magnetic separation and alkaline washing. In their study, the recovered iron effectively removed dissolved sulfide from sewage. It should be noted that land application of biosolids containing excess iron should be carefully considered, as excess iron may pose toxic effects on plants (Okoro and Sun, 2019). Thus, utilizing waste iron sources or recovering iron from waste sludge can be a more environmentally sustainable approach for cost-effective biogas desulfurization. However, such ideas are still in the embryonic phase of development.

3.5. Operational challenges

Iron dosing can provide an easily retrofittable desulfurization solution, as it can be implemented with a small capital investment (e.g., chemical dosing pumps). While demonstrating effective desulphurization efficiencies, various iron compounds and dosages have been utilized. Thus, the results cannot be generalized, and a site-specific optimization would be essential to avoid any operational disturbances of digesters. For instance, depending on the residence time and sulfide levels, iron salts may impede the bioavailability of phosphates due to co-precipitation, which may negatively affect methanogenic activity (Zhang et al., 2009). The co-precipitation of other compounds would also inevitably increase the required dosages and operating costs to attain target desulfurization efficiency. Thus, a holistic process optimization considering digester operating parameters (e.g., residence time) would be necessary for implementing iron dosing.

Interestingly, previous studies provided limited information on process operation or dosing strategy. Cao et al. (2019) highlighted that different dosing strategy might lead to different desulfurization efficiencies. Their study demonstrated that a low-dosage, high-frequency (i.e., interval between two dosages) Fe (III) dosing could provide better sulfide removal from sediments of gravity sewers. Nonetheless, reports on similar optimization could not be found for AD. Although it was not reported in previous studies, potential clogging by iron precipitates should not be overlooked. For instance, at the beginning of a full-scale demonstration (Erdirencelebi and Kucukhemek, 2018), iron dosing was kept at a low level to avoid any clogging.

4. Economic feasibility of in-situ technologies

Despite various in-situ desulfurization technologies often referred to as cost-effective, there has been a limited number of published reports on economic feasibility and systematic economic-based comparison of different in-situ and ex-situ technologies. In 2019, Okoro and Sun (2019) published a comprehensive systematic review on economic considerations and quantitative assessment/comparison of biogas desulfurization cost (US$/$m^3$ of biogas), annualized capital, and operating costs for two major in-situ desulfurization methods (i.e., microaeration and iron dosing) and major ex-situ (post-AD) desulfurization technologies (scrubbing, biofilter, biotrickling filters, absorption, and adsorption). Their analysis included inflation considerations, and it has been assumed that operating costing reported in the literature did not change significantly over the years. The authors performed uncertainty analysis for 50–150 % variations in the annual operating cost and the annualized capital costs. The detailed methodology and literature data used for the economic assessment can be found in their article (Okoro and Sun, 2019). Their economic assessment clearly indicated that in-situ technologies (microaeration and chemical (iron) dosing) could potentially provide a much cheaper solution than three ex-situ technologies (biotrickling filters, absorption, and adsorption). For instance, absorption using alkaline solution (NaOH) was the most expensive method (median cost of US $2.38/$m^3$). In general, ex-situ physicochemical desulfurization processes can lead to higher unit desulfurization costs due to several reasons. The high chemical cost for absorption (NaOH cannot be regenerated), and regular replacement of the cost of adsorbents (due to clogging issues) can lead to high unit desulfurization costs. Interestingly, the biotrickling filter also showed a high unit desulfurization cost (US $1.52/$m^3$). The authors suggested that such a high unit desulfurization cost could be attributed to the nutrient replenishment and supply through auxiliary trickling systems. The estimated median unit desulphurization costs of chemical (iron) dosing and microaeration were US $0.01/$m^3$ and US $0.016/$m^3$, respectively. Thus, between two in-situ desulfurization technologies, iron dosing was identified as relatively more cost-effective. The higher cost of microaeration resulted from the additional cost associated with oxygen and air supply systems (e.g., aeration devices, and pumps). However, it must be noted that previous reports suggested that microaeration could enhance biogas productivity due to the improved hydrolysis and development of a robust microbiome (Chen et al., 2020; Nguyen and Khanal, 2018).

Furthermore, Yang and Deng (2020) reported that microaeration could decrease the mixing requirement in an anaerobic digester, while mixing represents a significant portion (14–54 %) of total energy demand in AD facilities (Kariyama et al., 2018). Such possibilities are not considered in the economic assessment by Okoro and Sun (2019). Furthermore, the literature data used for their economic analysis considered the average costing data from oxygen and pure air-based microaeration (Díaz et al., 2015). Notably, the operating cost of oxygen-based microaeration can often be higher than that of an air-based system, depending on the type of system used (Díaz et al., 2015). For instance, the utilization of concentrated oxygen (92–98 %) showed a higher net present value (NPV5 and NPV20) than pure oxygen or air utilization for microaerobic desulfurization. Therefore, we cannot rule out that considering a specific air/oxygen supply system may lead to different outcomes. At the same time, maintaining a continuous supply of chemicals (iron) can often be challenging.

A few studies also reported the economic feasibility of microaeration, compared to ex-situ iron sponge technology (Díaz et al., 2015; Okoro and Sun, 2019). These studies suggested that microaeration would be economically more feasible than iron-sponge desulfurization. In 2020, Huertas et al. (2020) compared the economic feasibilities of these two technologies for a small-scale field AD (8.5 m^3) in Peru. Their results showed that the desulfurization cost with microaeration would be 13–14 times lower than iron-sponge technology. Briefly, iron-sponge is one of the oldest ex-situ technologies for H_2S and mercaptans removal from gaseous medium (Cherosky and Li, 2013; Huertas et al., 2020). The iron-sponge process was originally developed for natural gas streams and used in Europe and the USA for over 100 years (Huertas et al., 2020). In this process, ferric oxide/hydroxide is impregnated onto sulfide adsorption media, such as wood chips or wood shavings (Cherosky and Li, 2013). One of the major advantages of this process is the regeneration of iron oxide through the formation of elemental sulfur by exposing the media to oxygen. However, the regeneration can be done a limited number of times until saturation occurs (Ghimire et al., 2021; Huertas et al., 2020). Therefore, the adsorption media need to be replaced intermittently. Moreover, the regeneration process is highly exothermic, thus posing a high risk of fire (Ghimire et al., 2021).

Thus, both iron dosing and microaeration can potentially provide cost-effective solutions for in-situ desulfurization compared to certain ex-situ technologies. However, the information can only guide to support preliminary decisions regarding the feasibility, and estimated costs reported in the literature should be carefully considered for any major financial decisions. Notably, estimated desulfurization costs may be influenced by many site-specific factors, such as H_2S levels. For instance, Okoro and Sun (2019) estimated unit desulfurization cost with microaeration process as US $0.016/$m^3$, while other studies suggested 4–8 times lower unit desulfurization cost (Díaz et al., 2015; Khoshnevisan et al., 2017). Therefore, a detailed site-specific techno-economic

assessment is recommended. Moreover, environmental considerations should be considered in selecting the most sustainable desulphurization technology.

5. Summary and key knowledge gaps

As evident from the literature, microaeration and iron dosing have proven to be effective methods for in-situ desulfurization of biogas during AD. In addition to the successful full-scale demonstration, both methods could provide up to 99 % H_2S removal from biogas. The following specific take-home messages and knowledge gaps can be summarized from the literature:

- Microaeration can be applied in either the liquid or gas phase, and both schemes exhibited effective desulfurization during full-scale demonstrations. However, gas-phase microaeration (i.e., injecting air/oxygen in headspace) was mostly investigated, possibly because liquid-phase microaeration may trigger more non-specific oxidation of organics, leading to lower methane yields.
- Microaeration can be operated with either air or pure oxygen. Under equivalent oxygenation, both can provide similar desulfurization efficiencies. However, utilization of air would cause dilution of biogas with N_2, and biogas post-processing may be required. Therefore, utilizing pure oxygen was often recommended; however, an effective process control strategy would still be critical to avoid residual O_2 in biogas. Considering an ORP-based oxygen dosing strategy seems to be a feasible approach. However, a closer look at the literature suggested inconsistencies between the theoretical ORP range for microaerobic conditions and optimum ORP values reported for microaeration. Thus, site-specific optimization would be necessary.
- Maintenance requirements, particularly intermittent removal of elemental sulfur from the digester headspace walls, would be necessary for microaeration operation. Based on reports, the disposition of elemental sulfur on digester headspace would be unavoidable even with liquid-phase microaeration. However, limited information is available on required cleaning frequency.
- As low oxygen levels are maintained, explosion/fire risks due to mixing of biogas/O_2 and inhibition of methanogenic activity do not appear to be a concern for microaeration operation.
- Various forms of iron have been used for in-situ desulfurization, and different iron compounds may provide different results. However, Fe (III) was endorsed by several researchers due to faster reaction/response and effectiveness within a wide range of pH. Furthermore, Fe(III) could provide faster reduction to Fe(II) with parallel oxidation of sulfides.
- In addition to direct dosing into the digesters, iron dosing in sewer systems also showed a positive impact on removing H_2S from biogas in downstream AD.
- Interestingly, there is far less information available on comprehensive process optimization for iron dosing. For instance, very few reports suggested that the frequency of iron dosing would also be critical for effective H_2S control, while most studies primarily focused on dosages optimization.
- Compared to microaeration, iron dosing would be more easily retrofittable with a chemical dosing pump. Microaeration would require more capital investment, such as oxygen generation units (when pure oxygen is used) and aeration devices. Furthermore, adopting headspace microaeration may require structural/design modification of digesters.
- Both iron and oxygen dosing requirements can be theoretically estimated based on the dissolved sulfide concentrations in influent. However, providing theoretical dosages may not be adequate due to the utilization of iron and oxygen via co-precipitation of other compounds and non-specific oxidation of organics, respectively.
- Although most emphasis has been given to desulfurization efficiencies, both processes can provide some additional benefits. Microaeration can enhance AD process kinetics and stability, and iron dosing can improve sludge dewaterability due to its coagulating ability. Also, iron can increase biomethane yield by enhancing enzymatic activity. However, excess dosing of iron and oxygen may lead to process disturbance. Increased oxygen levels may inhibit methanogenic activity. Iron can reduce the bioavailability of phosphorus via precipitation. However, process disturbance by iron dosing and microaeration has been rarely reported.
- Limited data exist on the economic feasibility of both techniques. Based on extensive literature search, only one study could be found that provided a quantitative financial (capital and operating costs) comparison of different desulfurization technologies (Okoro and Sun, 2019). The estimations were based on data previously reported in the literature. Thus, the results were essentially indicative and should be carefully used in making financial decisions. Nonetheless, it seems that both iron dosing and microaeration can provide cheaper desulfurization options than most ex-situ desulfurization methods (adsorption, absorption, and biotrickling filter). Between microaeration and iron dosing, iron dosing showed slightly lower capital and operating costs per unit biogas desulfurization. Notably, the additional cost of aeration systems and maintenance (elemental sulfur removal) required for microaeration led to such a difference.
- It should be noted that available full-scale demonstration reports on iron dosing suggested slightly lower desulfurization efficiencies than microaeration (68–99 % vs. 35–72 %) (see Tables 1 and 2). However, both processes could achieve complete desulfurization (~99 % H_2S removal) in lab/pilot studies. Thus, site-specific process optimization would be necessary for both methods, which may influence the operating costs. Therefore, site-specific lab/pilot studies and further economic assessment should be considered before adopting a specific method.
- For both processes, desulfurization efficiencies reported in the literature were primarily based on H_2S removal from biogas. The fate of other volatile sulfur compounds (e.g., mercaptans) and impurities (e.g., siloxane) has rarely been reported. Both microaeration and iron dosing rely on the chemical/biochemical removal of dissolved sulfide, while volatile organosulfur compounds like mercaptans are produced from proteins rather than dissolved sulfide (Novak et al., 2002). Despite the lack of data on mercaptans removal, it can be inferred that removal of these volatile organosulfur compounds is not guaranteed with these techniques.

6. Conclusions

Previous studies provided essential evidence on the effectiveness of microaeration and iron dosing for in-situ biogas desulfurization during AD through different mechanisms. However, further research is still needed for process control and optimization, including dosing intervals and optimal dosage. The maintenance requirements and process disturbance by iron dosing and microaeration have rarely been reported. Moreover, the fate of organosulfur compounds (e.g., mercaptans) and other impurities (e.g., siloxanes) during microaeration and iron dosing have not been investigated adequately. Finally, a holistic techno-economic assessment of microaeration and iron dosing is essential to ensure the economic sustainability of in-situ technologies.

CRediT authorship contribution statement

Seyed Mohammad Mirsoleimani Azizi: Investigation, Visualization, Writing – original draft, Writing – review & editing. **Basem S. Zakaria:** Investigation, Writing – review & editing. **Nervana Haffiez:** Investigation, Writing – review & editing. **Parisa Niknejad:** Investigation, Writing – review & editing. **Bipro Ranjan Dhar:** Conceptualization, Visualization, Writing – original draft, Writing – review & editing,

Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no conflict of interests or personal relationships that could influence this research work.

Data availability

Data will be made available on request.

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References

Akgul, et al., 2017. Assessing iron and aluminum-based coagulants for odour and pathogen reductions in sludge digesters and enhanced digestate dewaterability. Sci. Total Environ. 598, 881–888. https://doi.org/10.1016/j.scitotenv.2017.04.141.

Andreides, et al., 2021a. Stirring-based control strategy for microaerobic H_2S removal in sequencing batch anaerobic digesters. Fuel 306 (15), 121696. https://doi.org/10.1016/j.fuel.2021.121696.

Andreides, et al., 2021b. Key parameters influencing hydrogen sulfide removal in microaerobic sequencing batch reactor. Biochem. Eng. J. 168, 107951 https://doi.org/10.1016/j.bej.2021.107951.

Andriamanohiarisoamanana, F.J., Shirai, T., Yamashiro, T., Yasui, S., Iwasaki, M., Ihara, I., Nishida, T., Tangtaweewipat, S., Umetsu, K., 2018. Valorizing waste iron powder in biogas production: Hydrogen sulfide control and process performances. J. Environ. Manag. 208, 134–141. https://doi.org/10.1016/j.jenvman.2017.12.012.

Andriani, D., Rajani, A., Kusnadi, Santosa, A., Saepudin, A., Wresta, A., Atmaja, T.D., 2020. A review on biogas purification through hydrogen sulphide removal. IOP Conference Series: Earth and Environmental Science 483, 12034. https://doi.org/10.1088/1755-1315/483/1/012034.

Azizi, S.M.M., Haffiez, N., Zakaria, B.S., Dhar, B.R., 2022. Thermal Hydrolysis of Sludge Counteracts Polystyrene Nanoplastics-Induced stress during Anaerobic Digestion. ACS EST Eng. 2, 1306–1315. https://doi.org/10.1021/acsestengg.1c00460.

Baek, G., Kim, J., Lee, C., 2019. A review of the effects of iron compounds on methanogenesis in anaerobic environments. Renew. Sust. Energ. Rev. 113, 109282 https://doi.org/10.1016/j.rser.2019.109282.

Barua, S., Dhar, B.R.B.R., 2017. Advances towards understanding and engineering direct interspecies electron transfer in anaerobic digestion. Bioresour. Technol. 244, 698–707. https://doi.org/10.1016/j.biortech.2017.08.023.

Buisman, C.J.N., Geraats, B.G., Ijspeert, P., Lettinga, G., 1990. Optimization of Sulphur production in a biotechnological sulphide-removing reactor. Biotechnol. Bioeng. 35, 50–56. https://doi.org/10.1002/bit.260350108.

Cai, Y., Zhao, X., Zhao, Y., Wang, H., Yuan, X., Zhu, W., Cui, Z., Wang, X., 2018. Optimization of Fe2+ supplement in anaerobic digestion accounting for the Fe-bioavailability. Bioresour. Technol. 250, 163–170. https://doi.org/10.1016/j.biortech.2017.07.151.

Cao, J., Zhang, L., Hong, J., Sun, J., Jiang, F., 2019. Different ferric dosing strategies could result in different control mechanisms of sulfide and methane production in sediments of gravity sewers. Water Res. 164, 114914 https://doi.org/10.1016/j.watres.2019.114914.

Charles, W., Cord-Ruwisch, R., Ho, G., Costa, M., Spencer, P., 2006. Solutions to a combined problem of excessive hydrogen sulfide in biogas and struvite scaling. Water Sci. Technol. 53, 203–211. https://doi.org/10.2166/wst.2006.198.

Chen, Q., Wu, W., Qi, D., Ding, Y., Zhao, Z., 2020. Review on microaeration-based anaerobic digestion: State of the art, challenges, and prospectives. Sci. Total Environ. 710, 136388 https://doi.org/10.1016/j.scitotenv.2019.136388.

Cherosky, P., Li, Y., 2013. Hydrogen sulfide removal from biogas by bio-based iron sponge. Biosyst. Eng. 114, 55–59. https://doi.org/10.1016/j.biosystemseng.2012.10.010.

Das, J., Ravishankar, H., Lens, P.N.L., 2022. Biological biogas purification: recent developments, challenges and future prospects. J. Environ. Manag. 304, 114198 https://doi.org/10.1016/j.jenvman.2021.114198.

Dhar, B.R., Elbeshbishy, E., Hafez, H., Nakhla, G., Ray, M.B., 2011a. Thermo-oxidative pretreatment of municipal waste activated sludge for volatile sulfur compounds removal and enhanced anaerobic digestion. Chem. Eng. J. 174, 166–174. https://doi.org/10.1016/j.cej.2011.08.070.

Dhar, B.R., Youssef, E., Nakhla, G., Ray, M.B., 2011b. Pretreatment of municipal waste activated sludge for volatile sulfur compounds control in anaerobic digestion. Bioresour. Technol. 102, 3776–3782. https://doi.org/10.1016/j.biortech.2010.12.020.

Dhar, B.R., Nakhla, G., Ray, M.B.M.B., 2012. Techno-economic evaluation of ultrasound and thermal pretreatments for enhanced anaerobic digestion of municipal waste activated sludge. Waste Manag. 32, 542–549. https://doi.org/10.1016/j.wasman.2011.10.007.

Díaz, et al., 2010. Performance evaluation of oxygen, air and nitrate for the microaerobic removal of hydrogen sulphide in biogas from sludge digestion. Bioresour. Technol. 101 (20), 7724–7730. https://doi.org/10.1016/j.biortech.2010.04.062.

Díaz, I., Lopes, A.C., Pérez, S.I., Fdz-Polanco, M., 2011. Determination of the optimal rate for the microaerobic treatment of several H2S concentrations in biogas from sludge digesters. Water Sci. Technol. 64, 233–238. https://doi.org/10.2166/wst.2011.648.

Díaz, I., Ramos, I., Fdz-Polanco, M., 2015. Economic analysis of microaerobic removal of H2S from biogas in full-scale sludge digesters. Bioresour. Technol. 192, 280–286. https://doi.org/10.1016/j.biortech.2015.05.048.

Dykstra, C.M., Pavlostathis, S.G., 2021. Hydrogen sulfide affects the performance of a methanogenic bioelectrochemical system used for biogas upgrading. Water Res. 200, 117268 https://doi.org/10.1016/j.watres.2021.117268.

Erdirencelebi, D., Kucukhemek, M., 2018. Control of hydrogen sulphide in full-scale anaerobic digesters using iron (Iii) chloride: performance, origin and effects. Water SA 44, 176–183. https://doi.org/10.4314/wsa.v44i2.04.

Farghali, M., Andriamanohiarisoamanana, F.J., Ahmed, M.M., Kotb, S., Yamamoto, Y., Iwasaki, M., Yamashiro, T., Umetsu, K., 2020. Prospects for biogas production and H2S control from the anaerobic digestion of cattle manure: the influence of microscale waste iron powder and iron oxide nanoparticles. Waste Manag. 101, 141–149. https://doi.org/10.1016/j.wasman.2019.10.003.

Firer, D., Friedler, E., Lahav, O., 2008. Control of sulfide in sewer systems by dosage of iron salts: comparison between theoretical and experimental results, and practical implications. Sci. Total Environ. 392, 145–156. https://doi.org/10.1016/j.scitotenv.2007.11.008.

Ganigue, R., Gutierrez, O., Rootsey, R., Yuan, Z., 2011. Chemical dosing for sulfide control in Australia: an industry survey. Water Res. 45, 6564–6574. https://doi.org/10.1016/j.watres.2011.09.054.

Ge, H., Zhang, L., Batstone, D.J., Keller, J., Yuan, Z., 2013. Impact of iron salt dosage to sewers on downstream anaerobic sludge digesters: sulfide control and methane production. J. Environ. Eng. 139, 594–601. https://doi.org/10.1061/(asce)ee.1943-7870.0000650.

Ghimire, A., Gyawali, R., Lens, P.N.L., Lohani, S.P., 2021. Technologies for removal of hydrogen sulfide (H2S) from biogas. In: Emerging Technologies and Biological Systems for Biogas Upgrading, pp. 295–320. https://doi.org/10.1016/b978-0-12-822808-1.00011-8.

Giordano, A., Di Capua, F., Esposito, G., Pirozzi, F., 2019. Long-term biogas desulfurization under different microaerobic conditions in full-scale thermophilic digesters co-digesting high-solid sewage sludge. Int. Biodeterior. Biodegrad. 142, 131–136. https://doi.org/10.1016/j.ibiod.2019.05.017.

Gutierrez, O., Park, D., Sharma, K.R., Yuan, Z., 2010. Iron salts dosage for sulfide control in sewers induces chemical phosphorus removal during wastewater treatment. Water Res. 44, 3467–3475. https://doi.org/10.1016/j.watres.2010.03.023.

Huertas, J.K., Quipuzco, L., Hassanein, A., Lansing, S., 2020. Comparing hydrogen sulfide removal efficiency in a field-scale digester using microaeration and iron filters. Energies 13, 4793. https://doi.org/10.3390/en13184793.

Jenicek, et al., 2008. Use of microaerobic conditions for the improvement of anaerobic digestion of solid wastes. Water Sci. Technol. 58 (7), 1491–1496. https://doi.org/10.2166/wst.2008.493.

Jenicek, et al., 2014. Improving products of anaerobic sludge digestion by microaeration. Water Sci. Technol. 69 (4), 803–809. https://doi.org/10.2166/wst.2013.779.

Jeníček, P., Horejš, J., Pokorná-Krayzelová, L., Bindzar, J., Bartáček, J., 2017. Simple biogas desulfurization by microaeration – full scale experience. Anaerobe 46, 41–45. https://doi.org/10.1016/j.anaerobe.2017.01.002.

Jiang, et al., 2017. Selection of in-situ desulfurizers for chicken manure biogas and prediction of dosage. Polish J. Environ. Stud. https://doi.org/10.15244/pjoes/64909.

Kariyama, I.D., Zhai, X., Wu, B., 2018. Influence of mixing on anaerobic digestion efficiency in stirred tank digesters: a review. Water Res. 143, 503–517. https://doi.org/10.1016/J.WATRES.2018.06.065.

Khanal, S.K., Huang, J.-C., 2006. Online oxygen control for sulfide oxidation in anaerobic treatment of high-sulfate wastewater. Water Environ. Res. 78, 397–408. https://doi.org/10.2175/106143006x98804.

Khanal, S.K., Huang, J.C., 2003. ORP-based oxygenation for sulfide control in anaerobic treatment of high-sulfate wastewater. Water Res. 37, 2053–2062. https://doi.org/10.1016/S0043-1354(02)00618-8.

Khoshnevisan, B., Tsapekos, P., Alfaro, N., Díaz, I., Fdz-Polanco, M., Rafiee, S., Angelidaki, I., 2017. A review on prospects and challenges of biological H2S removal from biogas with focus on biotrickling filtration and microaerobic desulfurization. Biofuel Res. J. 4, 741–750. https://doi.org/10.18331/BRJ2017.4.4.6.

Kiilerich, B., Nielsen, A.H., Vollertsen, J., 2018. Kinetics of sulfide precipitation with ferrous and ferric iron in wastewater. Water Sci. Technol. 78, 1071–1081. https://doi.org/10.2166/wst.2018.382.

Klok, J.B.M., de Graaff, M., van den Bosch, P.L.F., Boelee, N.C., Keesman, K.J., Janssen, A.J.H., 2013. A physiologically based kinetic model for bacterial sulfide oxidation. Water Res. 47, 483–492. https://doi.org/10.1016/j.watres.2012.09.021.

Kobayashi, T., Li, Y.Y., Kubota, K., Harada, H., Maeda, T., Yu, H.Q., 2012. Characterization of sulfide-oxidizing microbial mats developed inside a full-scale anaerobic digester employing biological desulfurization. Appl. Microbiol. Biotechnol. 93, 847–857. https://doi.org/10.1007/s00253-011-3445-6.

Krayzelova, L., Bartacek, J., Díaz, I., Jeison, D., Volcke, E.I.P., Jenicek, P., 2015. Microaeration for hydrogen sulfide removal during anaerobic treatment: a review. Rev. Environ. Sci. Biotechnol. 14, 703–725. https://doi.org/10.1007/s11157-015-9386-2.

Krayzelova, L., Bartacek, J., Kolesarova, N., Jenicek, P., 2014. Microaeration for hydrogen sulfide removal in UASB reactor. Bioresour. Technol. 172, 297–302. https://doi.org/10.1016/j.biortech.2014.09.056.

Kulandaivelu, J., Shrestha, S., Khan, W., Dwyer, J., Steward, A., Bell, L., Mcphee, P., Smith, P., Hu, S., Yuan, Z., Jiang, G., 2020. Full-scale investigation of ferrous dosing in sewers and a wastewater treatment plant for multiple benefits. Chemosphere 250, 126221. https://doi.org/10.1016/j.chemosphere.2020.126221.

Nägele, H.J., Steinbrenner, J., Hermanns, G., Holstein, V., Haag, N.L., Oechsner, H., 2017. Innovative additives for chemical desulphurisation in biogas processes: a comparative study on iron compound products. Biochem. Eng. J. 121, 181–187. https://doi.org/10.1016/j.bej.2017.01.006.

Ncube, A., Cocker, J., Ellis, D., Fiorentino, G., 2021. Biogas from source separated organic waste within a circular and life cycle perspective. A case study in Ontario, Canada. Environ. Sustain. Indic. 11, 100134 https://doi.org/10.1016/j.indic.2021.100134.

Nghiem, et al., 2014. Oxidation reduction potential as a parameter to regulate micro-oxygen injection into anaerobic digester for reducing hydrogen sulphide concentration in biogas. Bioresour. Technol. 173, 443–447. https://doi.org/10.1016/j.biortech.2014.09.052.

Nguyen, D., Khanal, S.K., 2018. A little breath of fresh air into an anaerobic system: how microaeration facilitates anaerobic digestion process. Biotechnol. Adv. 36, 1971–1983. https://doi.org/10.1016/j.biotechadv.2018.08.007.

Novak, J., Glindemann, D., Murthy, S.N., Gerwin, S.C., Peot, C., 2002. Mechanisms for generation and control of trimethyl amine and dimethyl disulfide from lime stabilized biosolids. In: Odors and Air Pollutants Conference. Water Environ. Fed., pp. 288–298. https://doi.org/10.2175/193864702785140528

Okoro, O.V., Sun, Z., 2019. Desulphurisation of biogas: a systematic qualitative and economic-based quantitative review of alternative strategies. ChemEngineering 3, 76. https://doi.org/10.3390/chemengineering3030076.

Qiang, H., Niu, Q., Chi, Y., Li, Y., 2013. Trace metals requirements for continuous thermophilic methane fermentation of high-solid food waste. Chem. Eng. J. 222, 330–336. https://doi.org/10.1016/j.cej.2013.02.076.

Ramos, I., Fdz-Polanco, M., 2013. The potential of oxygen to improve the stability of anaerobic reactors during unbalanced conditions: results from a pilot-scale digester treating sewage sludge. Bioresour. Technol. 140, 80–85. https://doi.org/10.1016/j.biortech.2013.04.066.

Ramos, I., Peña, M., Fdz-Polanco, M., 2014a. Where does the removal of H2S from biogas occur in microaerobic reactors? Bioresour. Technol. 166, 151–157. https://doi.org/10.1016/j.biortech.2014.05.058.

Ramos, I., Pérez, R., Fdz-Polanco, M., 2014b. The headspace of microaerobic reactors: Sulphide-oxidising population and the impact of cleaning on the efficiency of biogas desulphurisation. Bioresour. Technol. 158, 63–73. https://doi.org/10.1016/j.biortech.2014.02.001.

Ramos, et al., 2014c. Microaerobic digestion of sewage sludge on an industrial-pilot scale: The efficiency of biogas desulphurisation under different configurations and the impact of O2 on the microbial communities. Bioresour. Technol. 164, 338–346. https://doi.org/10.1016/j.biortech.2014.04.109.

Rebosura, M., Salehin, S., Pikaar, I., Kulandaivelu, J., Jiang, G., Keller, J., Sharma, K., Yuan, Z., 2020. Effects of in-sewer dosing of iron-rich drinking water sludge on wastewater collection and treatment systems. Water Res. 171, 115396 https://doi.org/10.1016/j.watres.2019.115396.

Rebosura, M., Salehin, S., Pikaar, I., Sun, X., Keller, J., Sharma, K., Yuan, Z., 2018. A comprehensive laboratory assessment of the effects of sewer-dosed iron salts on wastewater treatment processes. Water Res. 146, 109–117. https://doi.org/10.1016/j.watres.2018.09.021.

Rodríguez, E., Lopes, A., Fdz-Polanco, M., Stams, A.J.M., García-Encina, P.A., 2012. Molecular analysis of the biomass of a fluidized bed reactor treating synthetic vinasse at anaerobic and micro-aerobic conditions. Appl. Microbiol. Biotechnol. 93, 2181–2191. https://doi.org/10.1007/s00253-011-3529-3.

Roussel, J., Carliell-Marquet, C., 2016. Significance of vivianite precipitation on the mobility of iron in anaerobically digested sludge. Front. Environ. Sci. 4, 60. https://doi.org/10.3389/fenvs.2016.00060.

Ryue, J., Lin, L., Kakar, F.L., Elbeshbishy, E., Al-Mamun, A., Dhar, B.R., 2020. A critical review of conventional and emerging methods for improving process stability in thermophilic anaerobic digestion. Energy Sustain. Dev. 54, 72–84. https://doi.org/10.1016/j.esd.2019.11.001.

Salehin, S., Rebosura, M., Keller, J., Gernjak, W., Donose, B.C., Yuan, Z., Pikaar, I., 2020. Recovery of in-sewer dosed iron from digested sludge at downstream treatment plants and its reuse potential. Water Res. 174, 115627 https://doi.org/10.1016/j.watres.2020.115627.

Sarker, S., Lamb, J.J., Hjelme, D.R., Lien, K.M., 2018. Overview of recent progress towards in-situ biogas upgradation techniques. Fuel 226, 686–697. https://doi.org/10.1016/j.fuel.2018.04.021.

Song, et al., 2020. Air Supplement as a Stimulation Approach for the in Situ Desulfurization and Methanization Enhancement of Anaerobic Digestion of Chicken Manure. Energy and Fuels 34 (10), 12606–12615. https://doi.org/10.1021/acs.energyfuels.0c01724.

Sun, J., Pikaar, I., Sharma, K.R., Keller, J., Yuan, Z., 2015. Feasibility of sulfide control in sewers by reuse ofiron rich drinking water treatment sludge. Water Res. 71, 150–159. https://doi.org/10.1016/j.watres.2014.12.044.

Walker, S.B., Sun, D., Kidon, D., Siddiqui, A., Kuner, A., Fowler, M., Simakov, D.S.A., 2018. Upgrading biogas produced at dairy farms into renewable natural gas by methanation. Int. J. Energy Res. 42, 1714–1728. https://doi.org/10.1002/er.3981.

Yang, H., Deng, L., 2020. Using air instead of biogas for mixing and its effect on anaerobic digestion of animal wastewater with high suspended solids. Bioresour. Technol. 318, 124047 https://doi.org/10.1016/j.biortech.2020.124047.

Yoshimura, M., Nakano, Y., Yamashita, Y., Oho, T., Saito, T., Koga, T., 2000. Formation of methyl mercaptan from L-methionine by Porphyromonas gingivalis. Infect. Immun. 68, 6912–6916. https://doi.org/10.1128/IAI.68.12.6912-6916.2000.

Žák, M., Bendová, H., Friess, K., Bara, J.E., Izák, P., 2018. Single-step purification of raw biogas to biomethane quality by hollow fiber membranes without any pretreatment – an innovation in biogas upgrading. Sep. Purif. Technol. 203, 36–40. https://doi.org/10.1016/j.seppur.2018.04.024.

Zakaria, B.S., Lin, L., Chung, T., Dhar, B.R., 2020. An overview of complementary microbial electrochemical technologies for advancing anaerobic digestion. Advances in Bioenergy 5, 129–167. https://doi.org/10.1016/bs.aibe.2020.04.004.

Zhang, L., Keller, J., Yuan, Z., 2009. Inhibition of sulfate-reducing and methanogenic activities of anaerobic sewer biofilms by ferric iron dosing. Water Res. 43, 4123–4132. https://doi.org/10.1016/j.watres.2009.06.013.

Zhou, P., Meshref, M.N.A., Ranjan, B., 2021. Optimization of thermal hydrolysis process for enhancing anaerobic digestion in a wastewater treatment plant with existing primary sludge fermentation. Bioresour. Technol. 321, 124498 https://doi.org/10.1016/j.biortech.2020.124498.

Zhou, Q., Jiang, X., Li, X., Jiang, W., 2016. The control of H2S in biogas using iron ores as in situ desulfurizers during anaerobic digestion process. Appl. Microbiol. Biotechnol. 100, 8179–8189. https://doi.org/10.1007/s00253-016-7612-7.